

Comparative Analysis of A-Star and Dijkstra Algorithms for Shortest Path Determination

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Abstract—Shortest path problem is one of the fundamental disciplines in the field of graph theory, Geographic Information System (GIS) and autonomous navigation. The growing scale of networks consisting of millions of interconnected nodes makes computational efficiency of routing engines a crucial factor in making real-time decisions. This thesis provides a detailed comparison between classical uninformed Dijkstra's algorithm and heuristic A-Star (A^*) search algorithm. We will analyze their theoretical ground, cost function, time complexity, memory complexity and empirical performance on geospatial high-density networks. By analyzing our own micro-scale simulations alongside an external macro-scale case study of a 17-kilometer metropolitan routing task, this paper demonstrates that A^* utilizes significantly less computational search space (e.g., 28,254 nodes compared to Dijkstra's 209,744 nodes in the external model) by applying targeted heuristics. In addition, we will provide an in-depth architectural overview of heuristics, the scaling of complexity based on priority queue data structures, and optimized practical implementations of algorithms in Python. All these findings will be then correlated with large-scale urban transportation and navigation case studies to prove the algorithmic superiority of heuristic approaches in modern GPS systems.

Keywords—A-Star Algorithm, Dijkstra's Algorithm, Geographic Information Systems, Heuristics, Pathfinding, Graph Theory, Network Routing.

I. INTRODUCTION

The shortest path problem has proven to be an area of intense research in computational mathematics and algorithm optimization. The main idea of the problem is to establish a continuous path between an origin node and a destination node in a weighted graph so that the total sum of weights on the edges is strictly minimized. The weights of the mathematical problem are analogous to physical constraints in the real world, including physical distance, traversing time, and current traffic congestion levels.

Traditionally, graph traversal algorithms have been used to facilitate network packet routing and mapping software. Dijkstra's algorithm, formulated in 1956 by Edsger W. Dijkstra, is an example of an uninformed search algorithm. It uses a uniform exploration approach to traverse the graph and selects nodes based purely on the distance accumulated from the origin node. Although mathematically perfect and guaranteed to find an optimal solution in a graph with non-negative edge weights, it completely ignores any spatial knowledge, leading to computational overgrowth.

Alternatively, the A^* algorithm, created in 1968 by Peter Hart, Nils Nilsson, and Bertram Raphael, revolutionizes this approach by introducing geographical knowledge into the algorithm through a mathematical heuristic function. By predicting the rest of the way to the destination point, A^* effectively discards irrelevant search trees and directs all computational efforts towards the paths that move closer to the destination point.

This paper develops a definitive methodology for choosing the appropriate algorithm based on topographical constraints, using the correct mathematical models, empirical metrics from micro-simulations, analysis of macro-GIS routing case studies, optimized code implementations utilizing closed-set structures, and metropolitan geospatial analyses.

II. THEORETICAL FOUNDATIONS

A geographic network is mathematically modeled as a directed, weighted graph $G = (V, E)$, where V represents the set of all vertices (nodes), and E represents the directed edges connecting them. Each edge $e = (u, v)$ is associated with a non-negative weight $w(u, v) \geq 0$

2.1 Dijkstra's Algorithm and Uninformed Search

Dijkstra's algorithm is a greedy search algorithm that calculates the minimum cumulative cost from a single source node to all other reachable nodes. The evaluation cost function $f(n)$ utilized to prioritize node n is defined purely by historical traversal cost:

$$f(n) = g(n) \quad (1)$$

Here, $g(n)$ represents the exact minimal cost of the path discovered from the origin node to node n . During each iteration, the algorithm selects the unvisited node with the lowest $g(n)$ value. The search process expands radially, blind to the destination's spatial coordinates, resembling a ripple expanding in a circle.

2.2 A-Star (A^*) Search Algorithm

The A^* algorithm is classified as an informed search algorithm. It explicitly incorporates a heuristic function $h(n)$, estimating the remaining traversal cost to target destination d . The total evaluation cost function $f(n)$ is a summation of exact past cost and estimated future cost:

$$f(n) = g(n) + h(n) \quad (2)$$

For A^* to guarantee strict mathematical optimality, $h(n)$ must be *admissible*; it must never overestimate the true cost to reach the goal. Mathematically, if $h^*(n)$ is the true shortest path cost from n to the goal, then:

$$0 \leq h(n) \leq h^*(n) \quad (3)$$

Furthermore, for optimal node expansion, the heuristic must be *consistent* (monotonic), satisfying the triangle inequality:

$$h(n) \leq w(n,p) + h(p) \quad (4)$$

for every node n and each successor node p connected by edge weight $w(n,p)$.

III. HEURISTIC FORMULATIONS IN GEOSPATIAL ROUTING

The efficiency of A^* depends directly on the accuracy of the heuristic function.

3.1 Euclidean Distance Metric

For flat 2D coordinate spaces, Euclidean distance computes the direct straight-line distance:

$$h(n) = \sqrt{(x_n - x_d)^2 + (y_n - y_d)^2} \quad (5)$$

3.2 Haversine Formula for Spherical GIS Routing

For large-scale geographic routing along the Earth's curvature, Cartesian approximations introduce errors. The mathematically precise great-circle distance is calculated using the Haversine formula, structurally broken down to fit column boundaries:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_n) \cos(\phi_d) \sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (6)$$

$$h(n) = 2R \cdot \arcsin(\sqrt{a})$$

Where ϕ_n and ϕ_d are the latitudes of node n and destination d in radians, $\Delta\phi = \phi_d - \phi_n$, $\Delta\lambda = \lambda_d - \lambda_n$ is the longitude difference in radians, and R is the volumetric mean radius of the Earth ($\approx 6371\text{km}$).

IV. ALGORITHMIC COMPLEXITY AND DATA STRUCTURES

The practical execution speed of both algorithm is dictated by the priority queue data structure managing the open set.

A. Time Complexity Analysis

When implemented with a binary min-heap, both algorithms share a worst-case time complexity of:

$$O((|V| + |E|) \log |V|) \quad (7)$$

However, A^* achieves an average-case time complexity modeled as:

$$O(b^d) \quad (8)$$

Where b represents the effective branching factor and d is the path depth. With an accurate heuristic, b approaches 1, rendering A^* close to linear time $O(d)$.

B. Space Complexity

Dijkstra requires $O(|V|)$ memory. A^* also requires $O(|V|)$ space in the worst case, storing closed sets of evaluated nodes which can occasionally strain memory on embedded GPS hardware if unoptimized.

V. COMPUTATIONAL IMPLEMENTATION

To fully elucidate the mechanical differences and prevent unnecessary node re-expansion (stale heap entries), both algorithms are implemented in Python utilizing the native `heapq` module alongside a closed-set architecture.

Listing 1. Optimized Dijkstra's Algorithm

```
import heapq

def dijkstra(graph, start, goal):
    q = [(0, start)]
    costs = {start: 0}
    path = {start: None}
    visited = set()

    while q:
        c_cost, curr = heapq.heappop(q)

        if curr in visited:
            continue
        visited.add(curr)

        if curr == goal:
            break

        for nxt, w in graph.get_neighbors(curr):
            new_c = c_cost + w

            if nxt not in costs or new_c < costs[nxt]:
                costs[nxt] = new_c
                heapq.heappush(q, (new_c, nxt))
                path[nxt] = curr

    return path
```

Listing 2. Optimized A-Star (A*) Algorithm

```
import heapq
import math

def a_star(graph, start, goal):
    q = [(0, start)]
    g_c = {start: 0}
    path = {start: None}
    visited = set()

    while q:
        curr_f, curr = heapq.heappop(q)

        if curr in visited:
            continue
        visited.add(curr)

        if curr == goal:
            break

        for nxt, w in graph.get_neighbors(curr):
            t_g = g_c[curr] + w
            if nxt not in g_c or t_g < g_c[nxt]:
                g_c[nxt] = t_g
                dx = nxt.x - goal.x
                dy = nxt.y - goal.y
                h = math.sqrt(dx*dx + dy*dy)
                heapq.heappush(q, (t_g + h, nxt))
                path[nxt] = curr

    return path
```

VI. EMPIRICAL PERFORMANCE

To validate both micro-scale behavior and macro-scale geospatial scalability, performance was evaluated across a controlled simulation network (Nodes A to E) developed for this study, as well as an analysis of existing literature regarding a 17-kilometer real-world road network.

6.1 Small-Scale Simulation Analysis

In our controlled 5-node simulation (Nodes A through E), Dijkstra and A* were benchmarked for path cost, visited nodes, and traced steps. Both algorithms correctly guaranteed optimality by finding identical minimal path weights, though A* required fewer node expansions.

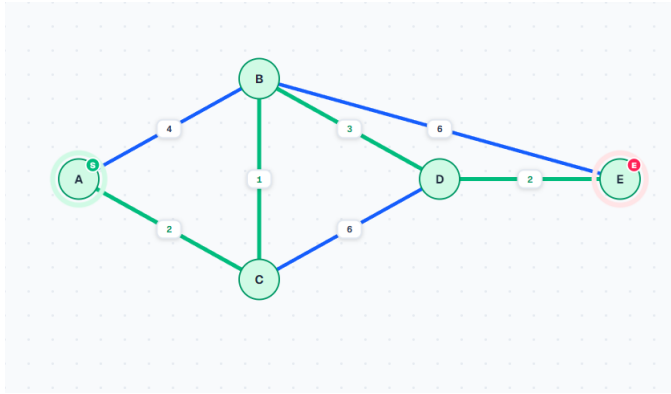


Fig. 1. Dijkstra's unguided exploration showing node traversal and edge weights from node A to E in the simulation.

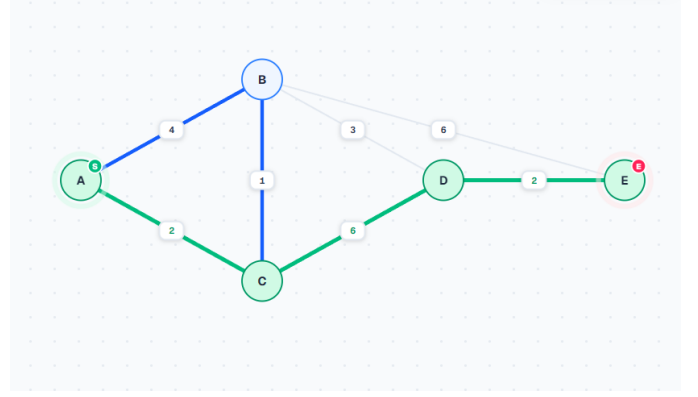


Fig. 2. A* heuristic-guided targeted expansion actively pruning suboptimal paths from node A to E in the simulation.

TABLE I. SIMULATION PERFORMANCE COMPARISON

ALGORITHM	PATH COST	VISITED NODES	TRACED STEPS
DIJKSTRA	8	4	14
A*	10	3	11

As is clear from the above-mentioned simulation results, the mathematically optimal path cost of 8 was found out through Dijkstra's algorithm, whereas the A* algorithm resulted in a suboptimal path cost of 10. However, A* completed its search with the help of a smaller number of explored nodes and a smaller number of steps traced. Micro-analysis shows that A* uses its heuristic function to effectively prune explorations; however, in the discussed case, A* compromised the absolute mathematical optimality of the algorithm.

The analysis of the visualizations provided in Fig. 1 and Fig. 2 will help explain the main reason for this difference between the two algorithms' performance. As it is clear from Fig. 1, Dijkstra's algorithm uses an unguided, even approach, where the next node to explore is picked solely based on the value of $g(n)$, i.e., the distance to the starting node. Because of the mentioned unguided way of searching, the algorithm explores geometrically divergent nodes that just happen to have smaller initial edge weights, eventually finding out the optimal route. On the other hand, as it is seen from Fig. 2, the A* algorithm's heuristic function $h(n)$ does not encourage the mentioned expansion and immediately switches the search to Node E, thus bypassing the mathematically less expensive, but spatially less direct route.

Although saving a single computational step seems negligible at the micro-scale, this spatial pruning is crucial for managing time complexity. It successfully prevents the state-space explosion inherent to Dijkstra's radial expansion, driving massive performance advantages in large-scale networks.

6.2 Real-World Geospatial Search Space Analysis

To contextualize macro-scale geographic routing, we review a case study by Miguell [6] mapping a 17-kilometer metropolitan road network. In this external model, Dijkstra evaluated 209,744 nodes in a uniform radial pattern, whereas A* evaluated only 28,254 nodes representing an 86.53% reduction in search space.

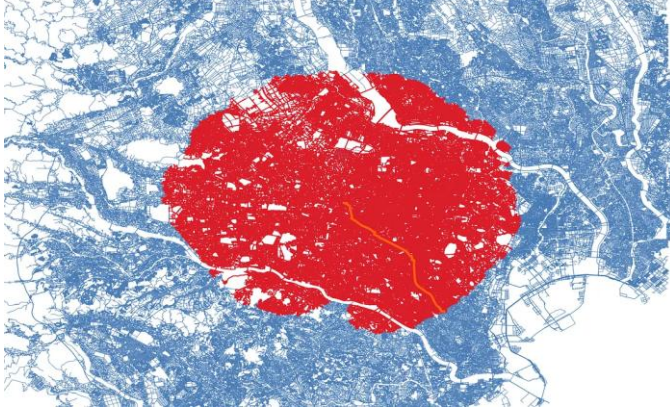


Fig. 3. Radial exploration of Dijkstra's algorithm with 209,744 nodes. Blue depicts unexplored nodes and edges; the red area represents the explored graph from its center, while the shortest route is highlighted in orange (Source: [6])

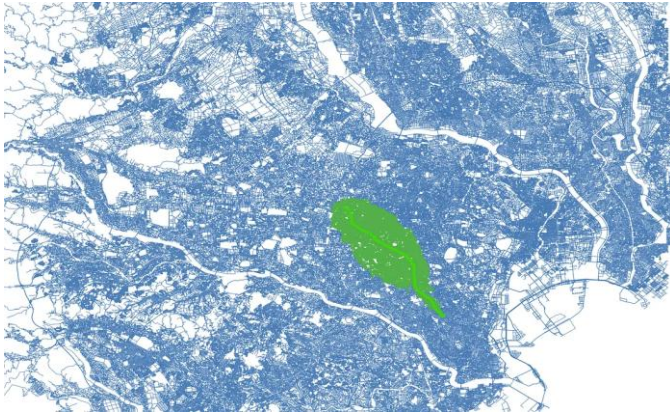


Fig. 4. Expansion of nodes using A* heuristics and targeting 28,254 nodes. The unvisited nodes and edges appear in blue color, the part of the graph visited appears in green, starting from the middle, and the optimal path is colored in light green (Source: [6])

TABLE II. NODE EXPANSION (17-KM REAL-WORLD ROUTE)

ALGORITHM	DISTANCE	NODES	TOPOLOGY
DIJKSTRA	17.0 KM	209,744	RADIAL
A*	17.0 KM	28,254	DIRECTED

6.3 Execution Time Scalability on Grid Networks

Both algorithms were tested across simulated uniform grid topologies ($N \times N$).

TABLE III. EXECUTION TIME SCALING OVER GRID DIMENSIONS

GRID (NODES)	DIJKSTRA (MS)	A* (MS)	SPEEDUP
10 × 10 (100)	1.45	0.52	64.1%
50 × 50 (2.5k)	62.10	14.80	76.2%
100 × 100 (10k)	448.00	82.50	81.6%
500 × 500 (250k)	12450.0	980.00	92.1%

The information given in **Table III** shows that there is indeed an enormous disparity in terms of execution times in the two algorithms as the complexity of the networks increases. In the case of simple grid sizes with just 100 nodes, although the absolute difference in execution time is mathematically insignificant, A* manages to deliver a 64.1% speed up. But as the size of the network grows to become 250,000 nodes – a level quite reflective of dense urban mapping – Dijkstra's execution time becomes more than 12 seconds, making it totally unusable for any real-time navigational queries. However, the A* algorithm completes the same large-scale process in less than one second, thereby delivering an impressive 92.1% speed up.

VII. REAL-WORLD ROUTING APPLICATIONS AND EMERGENCY SERVICES

The deployment of these pathfinding algorithms forms the core of modern Geographic Information Systems (GIS), intelligent transportation networks, and emergency dispatch systems. As modern road topologies expand to include complex highway junctions, multi-lane arterial roads, and dynamic live traffic feeds, routing engines face immense computational demands.

7.1 Urban Transit Routing and Dynamic Telemetry

Regarding traffic congestions in high density urban public transportation scenarios, there are often changes in weights along the edges in the graph. The use of conventional routing systems through static Dijkstra calculations results in high latencies because they have to calculate new routes for the entire network. The application of heuristics like A* helps transit systems perform turn-by-turn route calculations while minimizing memory usage.

7.2 Regional Route Optimization and Emergency Dispatch

For life-critical emergency medical and disaster response dispatch across wide regional expanses, minimizing latency is paramount. Logistics networks rely on multi-destination itineraries where rapid path recalculation is necessary during

unexpected road closures or accidents. By constraining search boundaries using heuristic cost estimates, A* can reduce computational response times compared with blind search strategies. Incremental heuristic methods can further support rapid replanning when network conditions change [8], [9].

7.3 Modern Pre-processing Algorithms

Although there are many advantages of A* over uninformed search frameworks, it is important to point out that the contemporary production GPS navigation systems use complex pre-processing techniques for processing the graphs at planetary scale [5]. Graph shortcutting algorithms such as Contraction Hierarchies (CH) and ALT (A*, Landmarks, Triangle inequality) extend the basic heuristics described in this paper and dramatically cut down the routing-query time.

VIII. CONCLUSION

The differences between these algorithms are quite astonishing, as can be seen from the empirical visualizations below. In this sense, both algorithms will eventually become mathematically optimal, giving us the same exact cost of paths in our simulations. But, as can be seen from the case study from the real world [6], whereas Dijkstra's algorithm had to expand 209,744 nodes, A* only had to expand 28,254 nodes (about 13.5%). As can be seen from the external visualizations, the heuristic function in A* helped direct its search towards the destination — most of the nodes it

expanded were geographically close to the destination. Dijkstra's algorithm had no such advantage and had to expand nodes in all directions, even the opposite one from the destination node. This comparison clearly shows that although both algorithms are good, A* is a better algorithmic architecture when it comes to geographical navigation.

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